

Lecture Notes on Set Theory

Unions, Intersections, Differences, Complements, and De Morgan's Laws

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1 Preliminaries: Notation, Quantifiers, Boolean Logic

Definition

Abbreviations and special notations (quick reference).

- **Membership:** $x \in A$ (“ x is in A ”), $x \notin A$.
- **Equality:** $A = B \iff (\forall x \in U) (x \in A \iff x \in B)$.
- **Subset:** $A \subseteq B \iff (\forall x \in U) (x \in A \Rightarrow x \in B)$.
- **Proper subset:** $A \subset B$ means $A \subseteq B$ and $A \neq B$.
- **Empty set:** $\emptyset$.
- **Universal set:** U (depends on context).
- **Operations:** $A \cap B$, $A \cup B$, $A \setminus B$, A^c .

Quantifiers and connectives:

- $\forall$ = “for all”, $\exists$ = “there exists”.
- $\Rightarrow$ implication, $\Leftrightarrow$ equivalence.
- Boolean: $\wedge$ (AND), $\vee$ (OR), $\neg$ (NOT).

Set membership as Boolean expressions:

$$x \in A \cap B \iff (x \in A) \wedge (x \in B), \quad x \in A \cup B \iff (x \in A) \vee (x \in B),$$

$$x \in A \setminus B \iff (x \in A) \wedge \neg(x \in B), \quad x \in A^c \iff \neg(x \in A).$$

2 Sets and Membership

Definition

Definition 2.1 (Set and membership). A **set** is a collection of objects called **elements**. If x is an element of A , we write $x \in A$; otherwise $x \notin A$.

Definition

Definition 2.2 (Universal set). In a given discussion, we fix a **universal set** U containing all objects under consideration. All sets we mention are assumed to be subsets of U .

Worked Example

Let $U = \{1, 2, 3, 4, 5, 6\}$ (outcomes of a fair die). Let $A = \{2, 4, 6\}$ (even outcomes). Then $2 \in A$ and $3 \notin A$.

Remark

Remark 2.1. Without specifying U , the word “complement” is ambiguous. Complements are always taken relative to the universal set U .

3 Subsets and Proper Subsets

Definition

Definition 3.1 (Subset). A is a **subset** of B (written $A \subseteq B$) if every element of A is also an element of B :

$$A \subseteq B \iff (\forall x \in U) (x \in A \Rightarrow x \in B).$$

Definition

Definition 3.2 (Proper subset). A is a **proper subset** of B (written $A \subset B$) if $A \subseteq B$ and $A \neq B$. Equivalently,

$$A \subset B \iff A \subseteq B \text{ and } \exists x \in B \text{ with } x \notin A.$$

Theorem / Proposition

Proposition 3.1 (Subset criterion). For sets $A, B \subseteq U$,

$$A \subseteq B \iff A \cap B^c = \emptyset.$$

Proof. $(\Rightarrow)$ If $A \subseteq B$ and $x \in A \cap B^c$, then $x \in A$ and $x \notin B$, contradicting $A \subseteq B$. Hence $A \cap B^c = \emptyset$.

$(\Leftarrow)$ If $A \cap B^c = \emptyset$ and $x \in A$, then $x \notin B$ would imply $x \in B^c$, hence $x \in A \cap B^c$, contradiction. Thus $x \in B$ and $A \subseteq B$. $\square$

Worked Example

Let $A = \{1, 2\}$ and $B = \{1, 2, 3, 4\}$. Then $A \subseteq B$. Since $3 \in B$ but $3 \notin A$, we have $A \subset B$.

4 Intersection

Definition

Definition 4.1 (Intersection).

$$A \cap B = \{x \in U : x \in A \text{ and } x \in B\}.$$

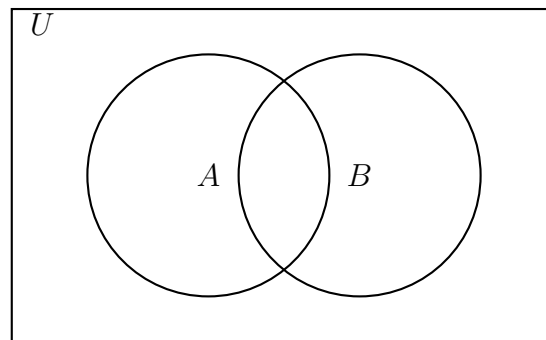


Figure 1: Venn diagram of $A \cap B$ (elements common to both sets).

Remark

Remark 4.1 (Boolean meaning).

$$x \in A \cap B \iff (x \in A) \wedge (x \in B).$$

Worked Example

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then

$$A \cap B = \{3, 4\}.$$

5 Union

Definition

Definition 5.1 (Union).

$$A \cup B = \{x \in U : x \in A \text{ or } x \in B\}.$$

Remark

Remark 5.1 (Boolean meaning).

$$x \in A \cup B \iff (x \in A) \vee (x \in B).$$

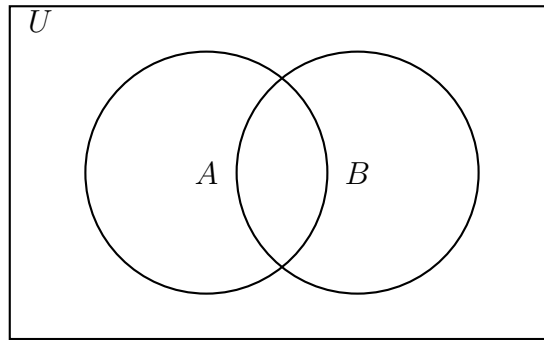


Figure 2: Venn diagram of $A \cup B$ (elements in A or in B or in both).

Worked Example

Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Then

$$A \cup B = \{1, 2, 3, 4, 5\}.$$

6 Set Difference

Definition

Definition 6.1 (Difference).

$$A \setminus B = \{x \in U : x \in A \text{ and } x \notin B\}.$$

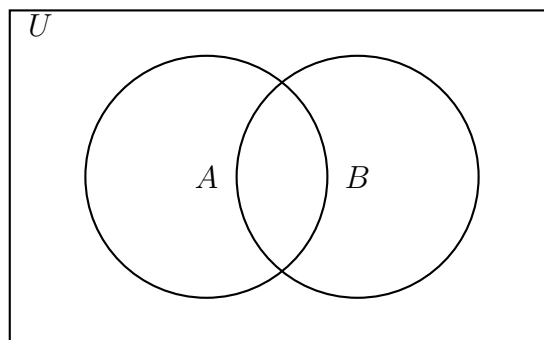


Figure 3: Venn diagram of $A \setminus B$ (elements in A but not in B).

Remark

Remark 6.1 (Boolean meaning).

$$x \in A \setminus B \iff (x \in A) \wedge \neg(x \in B).$$

Theorem / Proposition

Proposition 6.1 (Difference as intersection with complement).

$$A \setminus B = A \cap B^c.$$

Proof. For any $x \in U$,

$$x \in A \setminus B \iff (x \in A) \wedge (x \notin B) \iff (x \in A) \wedge (x \in B^c) \iff x \in A \cap B^c.$$

□

Worked Example

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$. Then

$$A \setminus B = \{1, 2\}, \quad B \setminus A = \{5\}.$$

7 Complements

Definition

Definition 7.1 (Complement). Let $A \subseteq U$. The complement of A (relative to U) is

$$A^c = U \setminus A = \{x \in U : x \notin A\}.$$

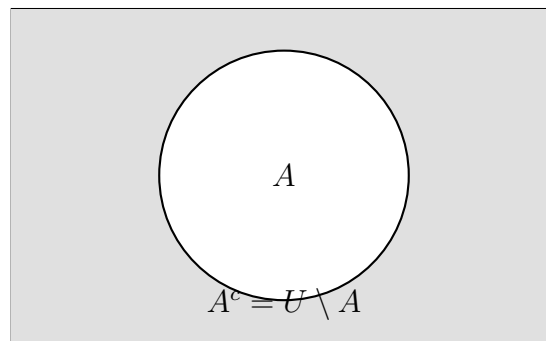


Figure 4: Venn diagram of A^c (everything in U but not in A).

Theorem / Proposition

Proposition 7.1 (Basic complement identities). *For any $A \subseteq U$,*

$$(A^c)^c = A, \quad A \cup A^c = U, \quad A \cap A^c = \emptyset.$$

Proof. Fix $x \in U$.

$$x \in (A^c)^c \iff x \notin A^c \iff \neg(x \notin A) \iff x \in A.$$

Also, each $x \in U$ satisfies $x \in A$ or $x \notin A$, hence $x \in A \cup A^c$, so $A \cup A^c = U$. Finally, $x \in A \cap A^c$ would mean $x \in A$ and $x \notin A$, impossible. $\square$

8 Core Algebra of Sets (Theorems & Proofs)

Theorem / Proposition

Proposition 8.1 (Idempotent laws).

$$A \cup A = A, \quad A \cap A = A.$$

Proof. For any $x \in U$,

$$\begin{aligned} x \in A \cup A &\iff (x \in A) \vee (x \in A) \\ &\iff x \in A, x \in A \cap A \iff (x \in A) \wedge (x \in A) \iff x \in A. \end{aligned}$$

$\square$

Theorem / Proposition

Proposition 8.2 (Commutative laws).

$$A \cup B = B \cup A, \quad A \cap B = B \cap A.$$

Proof. For any $x \in U$, OR and AND are commutative:

$$(x \in A) \vee (x \in B) \iff (x \in B) \vee (x \in A), \quad (x \in A) \wedge (x \in B) \iff (x \in B) \wedge (x \in A).$$

$\square$

Theorem / Proposition

Proposition 8.3 (Associative laws).

$$(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C).$$

Proof. Both follow from associativity of OR and AND on membership statements. $\square$

Theorem / Proposition

Theorem 8.1 (Distributive laws).

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \quad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Proof. Let $x \in U$. Then

$$\begin{aligned} x \in A \cap (B \cup C) &\iff (x \in A) \wedge ((x \in B) \vee (x \in C)) \\ &\iff ((x \in A) \wedge (x \in B)) \vee ((x \in A) \wedge (x \in C)) \iff x \in (A \cap B) \cup (A \cap C). \end{aligned}$$

The second identity is similar. $\square$

Corollary 8.1 (Difference via intersection and complement). *For any $A, B \subseteq U$,*

$$A \setminus B = A \cap B^c.$$

Remark

Remark 8.1. A standard proof technique in set theory is **element-chasing**: take an arbitrary $x \in U$ and translate membership into Boolean statements.

9 De Morgan's Laws

Theorem / Proposition

Theorem 9.1 (De Morgan's laws). *For any $A, B \subseteq U$,*

$$(A \cup B)^c = A^c \cap B^c, \quad (A \cap B)^c = A^c \cup B^c.$$

Proof. Let $x \in U$. Then

$$\begin{aligned} x \in (A \cup B)^c &\iff x \notin A \cup B \\ &\iff \neg((x \in A) \vee (x \in B)) \\ &\iff \neg(x \in A) \wedge \neg(x \in B) \\ &\iff x \in A^c \cap B^c. \end{aligned}$$

The second law is similar using $\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q)$. $\square$

Theorem / Proposition

Theorem 9.2 (De Morgan's Second Law). *For any sets $A, B \subseteq U$,*

$$(A \cap B)^c = A^c \cup B^c.$$

Proof. Let $x \in U$ be arbitrary. We show that x belongs to the left-hand side if and only if it belongs to the right-hand side.

$$x \in (A \cap B)^c$$

$$\iff x \notin A \cap B$$

$$\iff \neg((x \in A) \wedge (x \in B))$$

$$\iff \neg(x \in A) \vee \neg(x \in B) \quad (\text{by the logical equivalence } \neg(P \wedge Q) \iff (\neg P) \vee (\neg Q))$$

$$\iff (x \in A^c) \vee (x \in B^c)$$

$$\iff x \in A^c \cup B^c.$$

Since this holds for every $x \in U$, we conclude that

$$(A \cap B)^c = A^c \cup B^c.$$

□

10 Visual Demonstration of De Morgan's Law

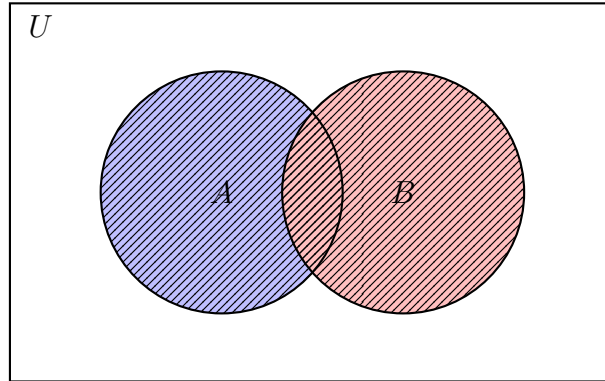


Figure 5: **(i) Region $A \cup B$** (black stripes mark the union).

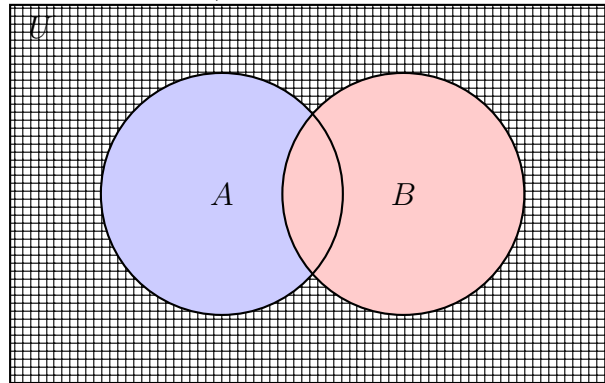


Figure 6: **(ii) Region $(A \cup B)^c$** (black grid outside both circles).

Color + pattern key: ■ = A , ■ = B , black stripes/grid = region being emphasized.

Demonstration:

- In diagram (i), the black striped region is *all points that lie in A or in B* . That is exactly the union $A \cup B$.
- In diagram (ii), we shade *everything in U* with a black grid, and then remove the union by making both circles white. So the remaining black grid region is *outside both circles*. That region is $(A \cup B)^c$.
- A point is outside both circles precisely when it is *not in A and not in B* . That means it lies in A^c and also in B^c , i.e. in $A^c \cap B^c$.

$$(A \cup B)^c = A^c \cap B^c.$$

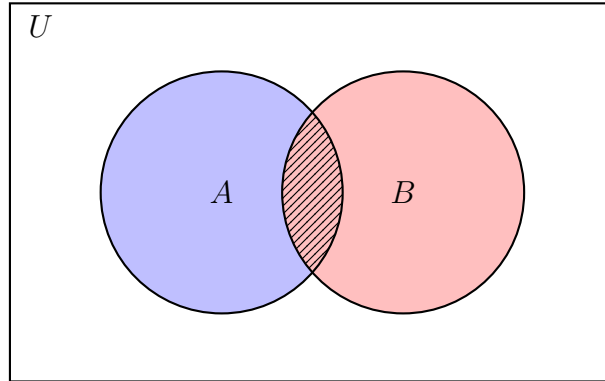


Figure 7: **(i) Region** $A \cap B$ (black stripes mark the overlap).

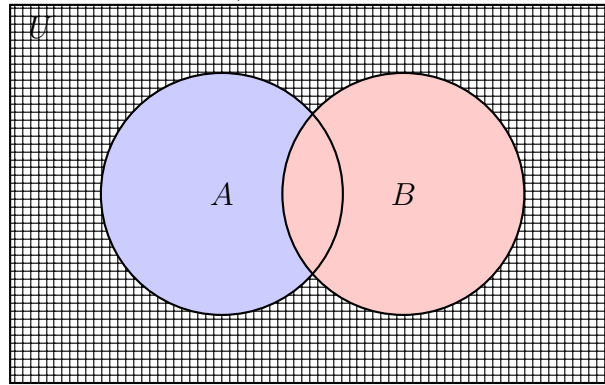


Figure 8: **(ii) Region** $(A \cap B)^c$ (black grid = everything except the overlap).

Color + pattern key: ■ = A , ■ = B , black stripes/grid = region being emphasized.

Demonstration:

- In diagram (i), the black striped region is the *overlap* of A and B . That region is $A \cap B$.
- In diagram (ii), we shade all of U with a black grid, then remove only the overlap (intersection) by painting it white. So the remaining black grid is *everything in U except the overlap*, i.e. $(A \cap B)^c$.
- Being outside the overlap means: *you fail to be in both sets simultaneously*. Equivalently, you are *not in A OR not in B* . That is exactly $A^c \cup B^c$.

$$(A \cap B)^c = A^c \cup B^c.$$